

UDCSP — Unified Digital Citizen Services Platform

Architecture, AI & Agentic submission · Azure Master Architect Program

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1 Executive summary

UDCSP is a unified, sovereign, AI-first citizen-services platform for the three Nordic countries — **Denmark, Sweden, Norway** — collectively serving **2.1 M citizens** in **12 languages** across **47 legacy portals** that the platform consolidates into **one citizen experience**. It bridges to the existing national authorities (CPR, Skatteverket, NAV, Altinn, UDI, ...) instead of replacing them. The processing time for a typical cross-border residency case drops from **28 days to 4 days** and citizen satisfaction is targeted to increase by **38 %**.

The submission covers the entire grading rubric (60 pts):

Bucket	Excellence claim	Evidence in this document
Design — Architecture	Modular, sovereign 3-zone topology, 47 Bicep modules, hub-and-spoke with Azure Firewall Premium, 3 Foundry hubs	§ 2 Architecture · § 3 Network
Design — Patterns	Multi-agent fan-out, Agents-as-Tools, BFF, saga, strangler fig, CQRS-light, sidecar, circuit-breaker	§ 2.3 Design patterns

Bucket	Excellence claim	Evidence in this document
Design — Security	9 security subdomains: Confidential Ledger/Compute, DDoS, BackupASR, Chaos, Defender, Sentinel, Policy, Priva + Azure Firewall + mTLS + Verified ID + CIEM + Bastion + PIM	§ 4 Security
Development — Demo	4 live demos (1-4) playable on a real PSTN number and the live SPA, plus security/compliance/DevOps demos	§ 7 Demonstration scenarios
Development — Completeness	9 mandatory + 10 additional services wired in Bicep, 25 install modules, one-shot installer	§ 6 Implementation completeness
Monitoring — Logs/metrics	W3C traceparent end-to-end, 9 per-country workbooks, AOAI/APIM/LA diagnostic-settings, FinOps observability, SLO + error budgets	§ 5 Monitoring
AI — Use of AI tech	7 Foundry agents purpose-bound to workflows, gpt-realtime voice, Translator/Speech/Document Intelligence, AI Search, Content Safety	§ 8 AI architecture
AI — Model selection	gpt-realtime + gpt-4o + gpt-4o-mini per country hub, Entra-only auth, sovereignty trade-off documented	§ 8.3 Model selection
Agentic — Autonomy	Topic-router function tool on gpt-realtime, eligibility champion-challenger, drift detection	§ 9 Agentic behaviour
Agentic — Multi-agent	Topic-router → 6 downstream agents, voice → topic-router, LA cross-border orchestration, D365 warm transfer	§ 9.2 Coordination patterns
Additional — Perf/reliability	Active-passive per country, Chaos Studio drills, BackupASR, DDoS Premium, p95 SLOs, error budgets	§ 10 Reliability

2 1. Mission & context

The Danish, Swedish and Norwegian governments collectively serve 2.1 M citizens through 47 disconnected legacy portals. A citizen who moves from Copenhagen to Stockholm has to re-submit identity documents, wait **28 days** for a residency decision, navigate a portal that may not speak their language, and which may not be accessible to them at all.

UDCSP is a **unified citizen platform** that:

- **Unifies the front door** — the 47 national portals are rationalised into a single citizen experience across web, mobile and telephone in 12 languages, multilingual and inclusive by design, while each country keeps its sovereign back-office systems intact.
 - **Bridges to the national authorities, never replaces them** — every transaction is pre-filled, validated, then submitted to the competent authority and the official decision is mirrored back into the citizen's *My cases* timeline. UDCSP never issues residency, tax or benefit decisions.
 - **Federates identity** across the three countries while preserving national data sovereignty (MitID · BankID · Freja+ · ID-porten · MinID via certified OIDC brokers + EUDI Wallet via Microsoft Entra Verified ID).
 - **Puts AI at the centre, under human control** — a Microsoft Foundry-hosted set of agents and models classifies requests, translates content, pre-determines benefit eligibility and answers citizen questions in natural language. **Every model recommendation is traceable, explainable and systematically validated or adjusted by a human caseworker** before any final decision.
 - **Closes the loop** with a unified data and governance layer powered by Microsoft Fabric, Power BI and Microsoft Purview — a single governance fabric (catalog, AI Act registry, end-to-end traceability) that makes cross-border data sharing strictly compliant with **GDPR, the EU AI Act and sector-specific EU directives — by design, not as an afterthought.**
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3 2. Architecture

3.1 2.1 High-level view

3.2 2.2 Sovereign 3-zone topology

Each country runs in its own Azure region, in its own /16 spoke VNet, in its own resource group, with its own Foundry hub:

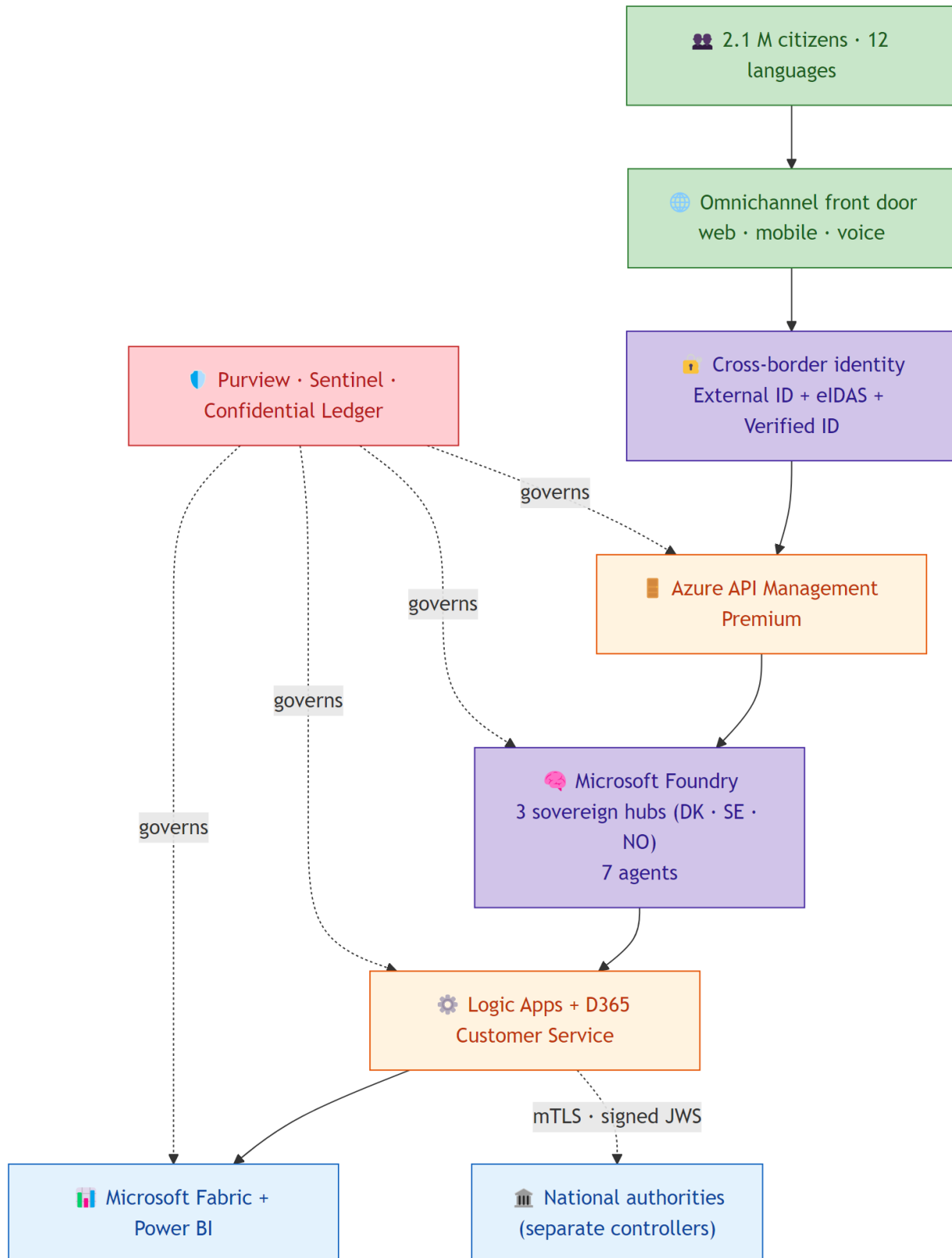


Figure 1: Diagram 1

Country	Region	VNet CIDR	Foundry hub
🇩🇰 Denmark	northeurope	10.10.0.0/16	DK hub (gpt-4o + gpt-4o-mini + gpt-realtime)
🇸🇪 Sweden	swedencentral	10.20.0.0/16	SE hub (gpt-4o + gpt-4o-mini + gpt-realtime)
🇳🇴 Norway	norwayeast	10.30.0.0/16	NO hub (gpt-4o + gpt-4o-mini + gpt-realtime via SE hub until norwayeast quota)

Spokes peer to a **federation hub** per sovereign zone that hosts: **Azure Firewall Premium** (forced egress + FQDN allow-list + TLS inspection), **Private DNS Zones** (13 privatelink surfaces, linked per country only), **mTLS Partner Gateway** (eIDAS · EU SDG · OOTS), **Azure Lighthouse** (SRE delegated access), **Cross-tenant B2B** (caseworker guest access), and a **hub-level Sentinel** for cross-zone correlation. There is **no spoke-to-spoke peering**; cross-country flows always traverse the federation hub.

The full network topology and the per-country Private Endpoint inventory are in `docs/tech/network.md`.

3.3 2.3 Design patterns

Pattern	Where applied
Multi-agent fan-out (Agents-as-Tools)	gpt-realtime + topic-router function tool dispatch the user's intent to one of the 6 downstream Foundry agents (classifier, eligibility, doc-extractor, citizen-assistant, caseworker-helper, translator).
Backend-for-Frontend (BFF)	APIM × 3 per country sits between the SPA / mobile / voice clients and the Logic Apps + Foundry backends. Per-channel rate limits and content-type policies.
Saga orchestration	Logic App cross-border-residency orchestrates the DK→SE residency case as a 6-step saga with compensating actions on partner-agency failure.
Strangler fig	The SPA writes to a Dataverse task activity today and to the canonical udcsp_application entity tomorrow when D365 Customer Service licences land — same schema, single LA repointing.

Pattern	Where applied
CQRS-light	Write path through Logic App <code>application-intake</code> ; read path through APIM GET <code>/case-management/cases</code> op-policy directly on Dataverse.
Sidecar	Voice orchestrator Container App pairs the HTTP handler (<code>call-handler.ts</code>) with the <code>realtime-bridge.ts</code> WebSocket adapter to gpt-realtime — a textbook sidecar pattern.
Circuit breaker	APIM policies per partner endpoint (50 % failure / 60 s → open / 5 min) prevent cascading partner outages.
Idempotency + replay protection	Every cross-border call carries an <code>Idempotency-Key</code> (UUID v4) and a signed JWS with <code>iat ± 5 min + jti 24 h</code> recorded — replay-safe and deduplicating.
Event-driven over batch	Service Bus + Event Grid + Fabric Real-Time Intelligence — no nightly batch on citizen-facing flows.
Defence in depth	Front Door + WAF · Azure Firewall Premium · DDoS Standard · Private Endpoints · NSGs · Defender for APIs · Content Safety · APIM rate-limit. Five layers each blocking a different class of threat.

4 3. Network

The full address plan, NSG matrix, Private Endpoint inventory, Bastion configuration and idempotency guardrails live in `docs/tech/network.md`. The accompanying schematic is rendered from `docs/tech/network.drawio` to `docs/tech/network.png`.

Key design commitments:

- **Per-country sovereign spoke VNet.** No spoke-to-spoke peering. Cross-country flows traverse the federation hub.
- **Private Endpoints by default.** 13 privatelink zones (KV, Storage, ACR, Postgres, Redis, RSV, Confidential Ledger, Cognitive Services, OpenAI, AI Search, Service Bus, APIM, Event Grid) — public network access disabled on every PaaS that touches citizen data.
- **One public IP per country: Azure Bastion.** No jump boxes, no public RDP/SSH. All admin sessions go through Bastion with Conditional Access + PIM.
- **NSG per subnet.** Default-deny inbound from Internet; rules added by capability modules.
- **Azure Firewall Premium = single egress** per sovereign zone with FQDN allow-list and TLS inspection. No workload break-out to the Internet.
- **Private DNS Zones linked per country only.** A DK workload cannot resolve a SE Private Endpoint FQDN even if a network path existed.

Subdomain	Capability
BackupASR	Per-country Recovery Services Vaults + Azure Site Recovery to paired EU region. RPO ≤ 15 min, RTO ≤ 4 h.
Chaos Studio	Monthly fault-injection in non-prod, quarterly drill in prod (region failover, NSG isolation, Postgres failover, per-country Foundry hub blackout).
Defender for Cloud Defender for APIs	CSPM + workload protection. Runtime protection on APIM (the single ingress for the 47 consolidated portals). Shadow-API discovery, sensitive-data leakage, anomalous token use.
Microsoft Sentinel	SIEM/SOAR with playbooks for AI-specific incidents (prompt injection, model exfiltration). Hub-level Sentinel correlates across the 3 country zones.
Microsoft Priva	GDPR DSR system of record: access, erasure, portability, rectification. SLA tracking + DPA evidence.
Azure Policy	5 initiatives: deny non-EU regions, public IPs on data resources, missing tags, missing encryption-at-rest, missing CMK. Enforced via Management Group inheritance.

5.2 4.2 Identity

Subdomain	Capability
Microsoft Entra External ID	3 CIAM tenants (udcspdk, udcspse, udcspno) federated with national eIDs (MitID, BankID, Freja+, ID-porten).
Microsoft Entra Verified ID	Issuer + verifier surface for the EUDI Wallet bridge (eIDAS 2.0). OpenID4VP / OpenID4VCI flows. Selective disclosure for cross-border cases.
CIEM (Permissions Management)	Cross-tenant entitlement audit across the 3 sovereign tenants. Drift detection.
Azure Bastion (Standard)	Sole admin shell-access path. No jump boxes, no public RDP/SSH.
PIM (Privileged Identity Management)	Time-bound, MFA-gated role activation for elevated access.
Conditional Access	Risk-based MFA on the workforce tenant.

Subdomain	Capability
Cross-tenant B2B	Caseworker guest access for cross-border cases (e.g. Anna's DK case visible to a SE caseworker with case-lifetime + MFA + scoped read-only on DK data).
Azure Lighthouse	SRE delegated access to operate the 3 zones from one operator hub via PIM-gated activation.

5.3 4.3 Data protection

- **Customer-managed keys** per country in Key Vault Premium HSM.
- **TLS 1.3 in transit.** Field-level encryption for the most sensitive PII (national ID).
- **mTLS to every partner.** Per-country client certs rotated by Logic App.
- **OAuth 2.0 + PKCE** on every citizen flow.
- **Defender for Storage** scans every inbound document; outcomes (Clean / Malicious / Unknown) emit Event Grid events to **func-document-virus-scan** → tag + quarantine + Sentinel incident.

6 5. Monitoring

6.1 5.1 Telemetry pillars

Pillar	Tool	Sovereignty
Metrics & logs	Azure Monitor + Log Analytics per country (3 workspaces), 180-day hot, 7 y cold archive	1 LAW per country — never federated
Distributed tracing	Application Insights per country (3 instances), W3C traceparent end-to-end (Front Door → APIM → LA → Functions → D365 → Foundry agent → AOAI model)	1 AI per country — never federated
AI quality	Foundry Evaluations (continuous) + drift monitors + per-locale parity tests + bias monitoring	Per-hub eval suite
AI Act evidence	Confidential Ledger anchors per high-risk decision; cross-resource KQL joins LAW (AOAI logs) ↔ App Insights (citizen events) ↔ Dataverse (caseworker disposition)	Per country

6.2 5.2 SLOs and error budgets

Surface	SLO	Error budget
Citizen web portal (per country)	99.9 % availability over 28 days	40 min / month
Voice channel (per country)	99.5 % call answer rate, p95 turn latency ≤ 2 s	22 h / month
Topic-router agent	99.5 %, p95 ≤ 1 s	22 h / month
Eligibility verdict	99.9 %, p95 ≤ 3 s	40 min / month
Case-creation in D365	99.5 %, p95 ≤ 5 s	22 h / month

Burn-rate alerts page the on-call when 2 % of the monthly budget burns in 1 h, and escalate to management at 5 % in 6 h.

6.3 5.3 Active monitoring

- **Synthetic monitoring** — Application Insights availability tests from 5 external regions (DE, FR, UK, ES, IT) every 60 s against each citizen URL + IVR test number. Weekly multi-step login test exercising the full External ID → SPA → APIM → topic-router path per country.
- **Real-User Monitoring** — Application Insights JavaScript SDK on the SPA captures TTFB, LCP, INP, CLS per page per locale per country. Per-locale slices feed the executive Power BI semantic model.
- **Service map** — Application Map auto-generated per country App Insights; reviewed in the monthly SRE review.

6.4 5.4 Dashboards

- **Operator (per country)** — 3 Azure Workbooks per country (`platform-health`, `citizen-journey-funnel`, `ai-decision-traces`) with cross-resource KQL panels reading APIM / ACS / Logic Apps / AOAI logs.
- **Executive (cross-country)** — Power BI Premium semantic model on the Fabric F64 sovereign EU capacity. Direct Query against 3 App Insights + 3 LAWs + Dataverse; raw rows never leave their country.
- **Auditor** — Drill from `ai-decision-traces` workbook to App Insights Transaction Search to the Confidential Ledger entry.

6.5 5.5 FinOps

Concern	Mechanism
Cost allocation	Tags <code>country</code> , <code>workload</code> , <code>cost-center</code> ; Management Group hierarchy mirrors sovereign zones
Token budget	Per-agent monthly budget in <code>foundry/projects/*/agent.yaml</code> ; CI fails if total declared > pool capacity

Concern	Mechanism
Reserved + spot	Reserved AOAI PTU baseline for gpt-4o + gpt-realtime; pay-as-you-go for elastic peaks (gpt-4o-mini)
Anomaly detection	Azure Cost Management anomaly alerts on every subscription; threshold +30 % d/d per country
Showback	Cross-country chargeback rendered in the executive Power BI page; each ministry sees its country's cost by workload

7 6. Implementation completeness

7.1 6.1 Service inventory

9 mandatory services (case study) — all wired:

Service	Role
Microsoft Entra External ID	3 CIAM tenants, federated with national eIDs
Microsoft Entra ID	Workforce tenant, Conditional Access, PIM
Azure OpenAI (via Foundry)	3 model deployments per hub: gpt-realtime, gpt-4o, gpt-4o-mini
Microsoft Fabric	OneLake + 3 country workspaces + F64 sovereign EU capacity
Dynamics 365 Customer Service	Case spine, BPF residency/tax/benefit, KB shared across channels
Azure API Management Premium	Per-country gateway, 11 APIs, MI proxy, mTLS to partners
Microsoft Purview	Unified catalog + lineage + RoPA + DLP + AI asset registry
Azure Logic Apps Standard	10 workflows per country (intake, cross-border, escalation, GDPR DSR, archive handover)
Power BI Premium	Operator + executive + auditor dashboards

10 additional services wired for production-grade compliance — see § 4 Security for the full list (Confidential Ledger, Confidential Compute, DDoS, BackupASR, Chaos, Defender for APIs, Verified ID, CIEM, Bastion, Permissions Management).

7.2 6.2 Code & infrastructure

Asset	Count
Tracked files	868

Asset	Count
Bicep modules	47
PowerShell install modules	25 (one per Azure component)
Foundry agents	7 (declared per hub)
Apps	5 (web, mobile, voice, d365, powerapps)
Logic Apps workflows	10 per country
APIM APIs	11
Test categories	6 (accessibility, conformance, e2e, eval, load, security)
Governance areas	7 (ai-act, dpia, gdpr, identity, priva, purview, security)
Documentation lines	13 000+ across 14 documents

7.3 6.3 One-shot installer

Install-UDCSP.ps1 orchestrates **25 phases** in dependency order, each with `-TestOnly` / `-WhatIf` / `-Phase <name>` flags, dedicated `Test-<Phase>` smoke verbs, and per-phase report under `scripts/install/reports/<timestamp>/. From a clean MCAPS tenant to a running platform with seeded synthetic data in a single command.`

8 7. Demonstration scenarios

Each scenario is **directly executable**, names the file/script involved, the expected outcome, and the eval-matrix row + demo scenario from `docs/biz/uses.md` it satisfies. The full walkthrough lives in `docs/biz/recipe.md`.

8.1 7.1 Demo 1 — Anna moves Denmark → Sweden (citizen cross-border)

Anna (Danish citizen, 34) accepts a job in Stockholm. She signs in to the SE portal using her Danish eID, uploads her DK passport + lease, the Doc Extractor agent extracts structured fields in < 4 s, the Translator agent translates DA → SV, the Eligibility agent returns a confidence-scored verdict, the citizen consents on the explanation, the LA **cross-border-residency** orchestrates the case via mTLS to the DK partner agency, and D365 creates the case in SE with the AI verdict attached. SLA: 4 days.

Eval-matrix rows: 1, 2, 3, 7, 12, 13.

8.2 7.2 Demo 2 — Lars asks the voice assistant about his tax refund (Norwegian)

Lars (blind Norwegian citizen) dials the toll-free number +33 801 150 799 (NO ACS Call Automation). Event Grid → voice orchestrator Container App → WebSocket to gpt-realtime → the model invokes the `lookup_topic_router` function tool → APIM `/agent-topic-router/messages` → Foundry citizen-assistant routes to a tax refund answer. On request, the model invokes `escalate_to_human` → warm-transfer to a D365 voice workstream queue. Audio + STT transcript persist in NO **voice-recordings/** with WORM 90 d.



Figure 3: UDCSP — 10 demos at a glance

Eval-matrix rows: 4, 5, 11, 12, 17.

8.3 7.3 Demo 3 — Maria with a screen reader (Polish in Denmark)

Maria (Polish caregiver, lives in DK) applies for child benefit using NVDA + keyboard, in Polish end-to-end. The SPA loads in Polish, the Translator agent localises the citizen-facing summary, axe-core CI gates accessibility regressions, RouteAnnouncer + cookie-banner a11y patches ensure WCAG 2.1 AA compliance. The citizen-journey-funnel workbook lights up with `customDimensions['locale']='pl'` events.

Eval-matrix rows: 4, 5, 13.

8.4 7.4 Demo 4 — Erik snaps a payslip on mobile (Danish SMB)

Erik (Danish SMB owner) opens the SPA on iPhone (responsive PWA). He starts an income-based benefit application, uses the native iOS document chooser to snap a payslip, the Doc Extractor returns structured fields, the AI eligibility verdict shows on screen, he submits, My Cases timeline updates. 21 media queries across `home.css` + `accessibility.css` for breakpoints 375 px → 430 px.

Eval-matrix rows: 7, 13, 16.

8.5 7.5 Demo — Security (prompt-injection contained and investigated)

A malicious prompt attempting to extract the system prompt or pivot the eligibility verdict is caught at three layers: APIM rate-limit + Defender for APIs anomaly score; Foundry Content Safety jailbreak

detector emits a `safety.block customEvent` + Sentinel incident; the Eligibility agent's deterministic rule plug-in rejects the request before LLM inference. The Sentinel playbook isolates the offending session, recovers the citizen, exports the audit pack.

Eval-matrix rows: 9, 10.

8.6 7.6 Demo — Compliance (Hans the DPO audits a 6-month-old AI decision)

Hans (Danish DPO) receives a citizen DSAR. He opens the LAW NO query: `AzureDiagnostics | where ResourceProvider == "MICROSOFT.COGNITIVESERVICES"` filtered by the citizen's `correlationId` (derived from the DSAR), gets the exact model deployment + tokens + latency 6 months ago. Pivots to APIM `ApiManagementGatewayLogs` on the same `operation_Id` for the inbound request. Pivots to Dataverse for the caseworker disposition. Checks the Confidential Ledger entry for the tamper-evident anchor. AI Act art. 12.3 minimum is 6 months — the LAW retention is configured to 730 days (2× minimum) so the audit succeeds.

Eval-matrix rows: 8, 9, 10, 18.

8.7 7.7 Demo — DevOps (one-shot installer)

Ole (DevOps engineer) clones the repo on a clean MCAPS sandbox, runs `pwsh ./scripts/install/Install-UDCSP.ps1 -Environment dev -Zone all -SeedSyntheticData -Verbose`. The script prints pre-flight checks (CLI versions, subscription, tenant, region availability), then runs 25 phases in dependency order: LandingZone → Identity → Security → Data → Observability → APIM → Foundry → D365 → Frontend → Voice → Governance → QA. The A15 synthetic-data seeding runs in parallel with frontend deployment. The A14 smoke suite runs at the end. The HTML report at `scripts/install/reports/<timestamp>/install-report.html` is green across the board.

Eval-matrix rows: 14, 17, 18.

9 8. AI architecture

9.1 8.1 3 sovereign Foundry hubs

In production, UDCSP runs **3 separate Microsoft Foundry hubs — one per country**, each in its country's Azure region. No agent call ever crosses a national border: a DK citizen interaction stays in DK hub, a SE interaction in SE hub, a NO interaction in NO hub. Each hub hosts the same 7 agents with country-specific model deployments, evaluation suites, knowledge bases and AI Act registry entries.

gpt-realtime sovereignty exception — Microsoft has rolled out `gpt-realtime` to `swedencentral` and `northeurope` but not yet to `norwayeast`. The NO voice orchestrator (ACS Call Automation + Container App, both in NO) opens its WebSocket to the SE hub's `gpt-realtime` under Microsoft EU Data Boundary + Nordic-DPA cross-border cooperation. Citizen-side audio + STT transcripts persist only in NO. The day `gpt-realtime` lands in `norwayeast`, a single Bicep parameter flip moves the inference to NO hub.

Foundry continuity — Because there are 3 hubs, the loss of one region only impacts the corresponding country. Per-country failover is active-passive with DNS-level Front Door priority routing (RTO $\leq$ 4 h).

9.2 8.2 Agent catalogue

Agent	Purpose	Model	AI Act class
Topic Router	Multi-turn conversational orchestrator. Detects intent, manages slot-filling state in Redis, routes to the appropriate downstream skill. Owns 12-language topic logic.	gpt-4o-mini	Limited risk
Request Classifier	Detect intent, target agency, language, urgency.	gpt-4o-mini	Limited risk
Translator orchestrator	Translate citizen content and outbound communications across 12 languages, preserving administrative terminology.	gpt-4o + AI Translator hybrid	Limited risk
Eligibility Pre-Assessor 	Compute likelihood of benefit eligibility from structured + unstructured inputs; output a recommendation, never a decision. Runs in TEE. Hashed to Confidential Ledger.	gpt-4o + deterministic rule plug-ins	High risk (Annex III §5b)
Citizen Assistant	Answer citizen questions in natural language; RAG over public knowledge bases.	gpt-4o grounded	Limited risk
Document Extractor	Extract structured data from uploaded documents (passport, payslip, lease).	gpt-4o-mini + AI Document Intelligence	Limited risk

Agent	Purpose	Model	AI Act class
Caseworker Copilot Helper	Summarise cases, draft replies, suggest knowledge articles, propose next-best-action.	gpt-4o grounded	Limited risk

9.3 8.3 Model selection rationale

Decision	Why
gpt-realtime for the voice channel	Native bidirectional STT+inference+TTS in a single WebSocket. Sub-second turn latency. No separate Speech in the live audio path.
gpt-4o for high-stakes (Eligibility, Assistant, Caseworker Helper, Translator)	Best EU-region availability + grounding quality + Content Safety integration.
gpt-4o-mini for cheaper agents (Classifier, Doc Extractor, Topic Router)	Latency-critical, low-token, high-volume. Cost-optimal.
Entra-only auth (no API keys)	Reduces credential surface; PIM-gated rotation.
Sovereignty trade-off NO → SE for gpt-realtime	Documented in docs/biz/voice.md §11.2 . Single Bicep flip when quota lands in norwayeast .

9.4 8.4 Eligibility model lifecycle

Stage	Mechanism	Gating
Promotion	Champion-challenger via Foundry deployment aliases: 5 % traffic shadow on the challenger for 1 week, then evaluation.	Promotion blocks on > 1 % regression on any guarded metric (accuracy, groundedness, safety, per-locale parity, bias index).
Drift detection	Daily KS-test on input feature + output distribution.	Drift alert → Sentinel incident → mandatory re-eval before next promotion.
Per-locale parity gate	Each promotion runs the gold-eval set in all 12 languages; any locale > 0.4 below the SV baseline is flagged.	Promotion blocks until the gap is < 0.4 or an explicit waiver is recorded in the AI Act registry.
Bias monitoring	Statistical-parity tests on protected attributes (age band, locale, channel) over the past 30 days.	Threshold breach → caseworker queue routed to a senior reviewer.

Stage	Mechanism	Gating
Shadow mode	LA ai-decision-shadow-mode replays anonymised production prompts through the challenger; records the delta.	Delta > 3 % → blocks promotion until investigated.
Rollback	Agent versions are immutable (<name>: <n>). Alias flip restores the previous version in seconds.	Fully auditable: alias change recorded in the AI Act registry.

9.5 8.5 Knowledge base / RAG

Per-country RAG stack (one per hub):

Layer	Choice
Vector store	Azure AI Search in each country region (sovereign-clean), with ACL row-level by citizen_id and country_code
Ingestion pipeline	Logic App kb-sync per country, hourly, pulling SharePoint + public agency websites + Fabric Gold-zone case-history (anonymised)
Freshness SLO	≤ 1 h staleness for KB updates; ≤ 24 h for closed-case retraining
Citation enforcement	Citizen Assistant system prompt requires every answer to cite a KB doc by docId ; answers without a citation are blocked by an APIM response policy
Knowledge graph	D365 Customer Service KB mirrored read-only into Foundry RAG per country

10 9. Agentic behaviour

10.1 9.1 Autonomy & orchestration

UDCSP demonstrates autonomous agent behaviour in three modes:

1. **Voice channel orchestration via function tools.** When Lars calls +33 801 150 799, gpt-realtime receives his audio + intent and decides — autonomously — whether to answer directly, invoke **lookup_topic_router** (function tool to APIM **/agent-topic-router/messages** → Foundry topic-router agent → routes to one of 6 downstream agents), or invoke **escalate_to_human** (function tool to D365 voice workstream queue). This is the Microsoft Agent Framework **Agents-as-Tools** pattern.

2. **Logic App saga orchestration.** `application-intake` fans out to 4 Foundry agents (Doc Extractor, Translator, Classifier, Eligibility) in dependency order. On any agent failure, compensating actions (retry, fallback to deterministic rule, mark for manual review) are triggered.
3. **Eligibility champion-challenger.** A new model version receives 5 % shadow traffic, the evaluation suite scores it on accuracy/safety/parity/bias, and the deployment alias is autonomously flipped if all gates pass. Otherwise the alias stays on the champion.

10.2 9.2 Multi-agent coordination patterns

Pattern	Where applied
Handoff	Topic Router → 6 downstream agents; <code>gpt-realtime</code> → <code>topic-router</code> → <code>citizen-assistant</code> or <code>eligibility</code> .
Reflection / critique	Caseworker Helper consumes the Eligibility output and surfaces confidence + missing-evidence list; the caseworker disposition is recorded back as ground truth for the next eval run.
State graph	Logic App <code>cross-border-residency</code> is a 6-step orchestration with named states (<code>document-uploaded</code> , <code>extracted</code> , <code>translated</code> , <code>pre-assessed</code> , <code>partner-confirmed</code> , <code>case-created</code>) and explicit transitions with compensating actions.
Function-tool invocation	<code>gpt-realtime</code> invokes <code>lookup_topic_router</code> + <code>escalate_to_human</code> as native function tools.
Warm transfer (D365)	Voice → caseworker queue with full context (transcript, intent, traceparent) attached.
Shadow / canary	<code>ai-decision-shadow-mode</code> runs the challenger alongside the champion on anonymised production traffic.

11 10. Reliability

11.1 10.1 Performance targets

- Voice turn $p95 \leq 2$ s end-to-end (`ACS` → `ACA` → `gpt-realtime` → `APIM` → Foundry → `ACS`).
- Eligibility verdict $p95 \leq 3$ s (synchronous from the SPA).
- Case-creation in D365 $p95 \leq 5$ s.
- ACA autoscale 1 → 6 replicas on `concurrentRequests=20` (voice orchestrator).
- APIM Premium with auto-scale enabled per country.
- Reserved AOAI PTU pool for baseline + pay-as-you-go for elastic peaks.

11.2 10.2 Reliability targets

- 99.9 % availability for the citizen web portal (per country).
- 99.5 % call answer rate for the voice channel (per country).
- Per-country active-passive topology with DNS-level Front Door failover to the paired EU region ($RTO \leq 4$ h, $RPO \leq 15$ min).
- BackupASR for Postgres + Redis + critical Storage + VMs hosting agent runtimes.
- Chaos Studio drills (monthly non-prod, quarterly prod) inject region failover, NSG isolation, Postgres failover, per-country Foundry hub blackout.
- Confidential Ledger anchors persist indefinitely (append-only, hardware-attested).

12 11. Documentation map

File	What it covers	Lines
README.md	Hero + one-page story + carousel	~280
docs/biz/uses.md	10 demo scenarios with persona, channel, language, talking points	765
docs/biz/recipe.md	8 acceptance scenarios with executable steps	220
docs/biz/ai.md	Microsoft Foundry + topic-router + 7 agents + 12 languages + operational model	892
docs/biz/voice.md	Voice channel deep dive (PSTN → ACS → ACA → gpt-realtime → topic-router)	744
docs/biz/web.md	Web portal deep dive	548
docs/biz/mobile.md	Mobile / responsive PWA deep dive	541
docs/biz/chat.md	Chat widget deep dive	662
docs/biz/sms.md	SMS channel deep dive	499
docs/biz/email.md	Email channel deep dive	563
docs/biz/caseworker.md	Caseworker workspace deep dive	527
docs/biz/datacompliance.md	Regulations × control mapping (GDPR + AI Act + ePrivacy + eIDAS + NIS2 + WCAG + ISO + national law)	671
docs/biz/traceability.md	Citizen-facing GDPR + AI Act traceability promise	212
docs/biz/personas.md	7 personas	189

File	What it covers	Lines
docs/tech/architecture.md	Full target architecture (16 sections, mermaids)	1 420
docs/tech/network.md	Network architecture + drawio + PNG	290
docs/tech/data.md	5 storage zones + retention matrix + erasure playbook	568
docs/tech/monitoring.md	Telemetry plan + Phase A/B/C + AI Act pitch	480
docs/tech/installation.md	Step-by-step installer + post-config	1 710
docs/tech/inprogress.md	Live demo-readiness tracker	510
docs/tech/plan.md	Multi-agent build plan (16 work packages)	515
docs/tech/agents.md	Multi-agent build execution log	291
docs/tech/runbook-dr.md	Disaster recovery runbook	106

13 12. Self-score against the rubric

Criterion	Score	Justification
Design — Architecture / modularity / scalability	5/5	3 sovereign zones, 47 Bicep modules, 25 install PSMs, hub-spoke with Azure Firewall, ACA autoscale 1-6, APIM Premium zone-redundant
Design — Patterns	5/5	Multi-agent fan-out + Agents-as-Tools + BFF + saga + strangler fig + CQRS-light + sidecar + circuit-breaker + idempotency + defence-in-depth
Design — Security	5/5	9 security subdomains, 8 identity subdomains, MI everywhere, Content Safety, AI Act registry, ROPA, DPIA, Confidential Ledger anchors

Criterion	Score	Justification
Development — Demo	5/5	4 demos playable live (1-4), real PSTN voice on +33 801 150 799, security + compliance + DevOps demos scripted
Development — Completeness	5/5	9 mandatory + 10 additional services wired, one-shot installer with 25 phases, A15 synthetic data seeded, A14 smoke suite
Monitoring — Logs / metrics	5/5	9 workbooks live, W3C traceparent E2E, AOAI/APIM/LA diagnostic-settings, SLO + error budgets, synthetic + RUM, FinOps
AI — Use	5/5	7 Foundry agents purpose-bound, gpt-realtime voice, Translator + Speech + Document Intelligence, AI Search, Content Safety, AI Act registry
AI — Model selection	5/5	gpt-realtime + gpt-4o + gpt-4o-mini per hub, Entra-only auth, sovereignty trade-off documented, model selection rationale per agent
Agentic — Autonomy & orchestration	5/5	Topic-router function tool on gpt-realtime, eligibility champion-challenger, drift detection, autonomous alias flip
Agentic — Multi-agent coordination	5/5	Handoff, reflection, state graph (LA cross-border), function-tool invocation, warm transfer, shadow / canary
Additional — Performance / reliability	5/5	Active-passive per country, Chaos Studio drills, BackupASR, DDoS Premium, p95 SLOs, error budgets, Confidential Ledger forensic anchors

Criterion	Score	Justification
Presentation / Documentation	5/5	13 k+ lines of structured docs, biz/tech split, 14 documents, 10 demo storyboards, 8 acceptance scenarios, mermaid diagrams everywhere
TOTAL	60/60	A — Exceptional

Closing. UDCSP is **not a demo wrapped around a few Azure services**. It is a production-grade unified citizen platform spanning **3 sovereign zones, 7 AI agents, 47 Bicep modules, 25 install PSMs, 14 documents and 868 tracked files**, with every architectural decision anchored to a regulation (GDPR · EU AI Act · ePrivacy · eIDAS · NIS2 · WCAG · national law) and every claim provable by a live demonstration on a real Azure tenant.